

Code-Layout, Readability and Reusability

Date	29 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the OptiCrop codebase in terms of its structure, readability, and reusability. The project is developed using Python, Flask, and Machine Learning libraries such as Pandas, NumPy, and Scikit-learn. Well-organized code with proper naming conventions, comments, modular design, and reusable functions ensures that the application is easy to maintain, extend, and reuse for future agricultural prediction and recommendation systems.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	PEP-8 followed
2	Proper File Structure	Files and folders are logically organized	Yes	Flask templates/static/models
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Crop/environment variables
4	Function / Method Names	Functions are descriptively named	Yes	predict_crop(), preprocess_input()
5	Code Comments	Inline and block comments explain logic	Partial	Important modules documented
6	Modular Design	Code is split into reusable functions/modules	Yes	ML, preprocessing, UI separated
7	No Redundant Code	Duplicate or unused code is removed	Yes	Common utilities reused

8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation added
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Reusable Components/Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High/Medium/Low)
1	Data Preprocessing	Python, Pandas	Training & Prediction	High
2	ML Prediction Model	Scikit-learn	Crop Recommendation	High
3	Flask Routes	Flask	Web Application	High
4	Input Validation	Python	Forms & Prediction	Medium
5	Visualization	Matplotlib/Seaborn	EDA & Reports	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized Flask project structure
Readability	5	Clear naming and formatting
Reusability	5	Reusable ML and preprocessing modules
Documentation / Comments	4	Adequate comments with scope for more
Overall Score	4.8/5	High-quality, maintainable codebase